

$^{32}\text{S}(\alpha, \text{d})$  1982De29, 1977Na10, 1976De24

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$^{32}\text{S}+1\text{p}+1\text{n}\rightarrow^{34}\text{Cl}$  from  $J^\pi=0^+$   $^{32}\text{S}$  ground state.

**1982De29, 1981De37:** a 50-MeV  $\alpha$  beam was produced from the KVI cyclotron. Targets were solid S evaporated onto C foil and covered with a C layer and gaseous  $\text{H}_2\text{S}$ . Reaction products were detected using a Si  $\Delta\text{E}$ -E telescope. Measured  $\sigma(\text{E}_\text{d}, \theta)$ . Deduced levels, L-transfers, J,  $\pi$ , and spectroscopic factors from zero-range DWUCK-DWBA analysis.

**1977Na10:** a 40-MeV  $\alpha$  beam was produced from the Michigan State University cyclotron. Targets were 99.9%-enriched  $^{32}\text{S}$  of  $110\text{ }\mu\text{g}/\text{cm}^2$  sandwiched between layers of Formvar and carbon foils. Reaction products were detected in the focal plane of an Enge split-pole magnetic spectrograph with a proportional-counter plastic-scintillator combination with FWHM=40-60 keV. Measured  $\sigma(\text{E}_\text{d}, \theta)$ . Deduced levels, J,  $\pi$ , and L-transfers.

**1976De24:** a 40-MeV  $\alpha$  beam was produced from the Princeton azimuthally varying field cyclotron. The target was CdS with natural  $^{32}\text{S}$ . Reaction products were detected using a freon cooled  $\Delta\text{E}$ -E Si detector telescope with FWHM=50 keV. Measured  $\sigma(\text{E}_\text{d}, \theta)$ . Deduced levels, L-transfers, J, and  $\pi$  from DWUCK-DWBA analysis. Also measured  $^{32}\text{S}(^3\text{He}, \text{p})^{34}\text{Cl}$ .

**1966Ki10:** a 29.0-MeV  $\alpha$  beam was produced from the 105-cm Kyoto University cyclotron. Targets were 0.78-mg/cm<sup>2</sup>-thick sulfur covered by thick gold foils. Reaction products were detected using a 50- $\mu\text{m}$   $\Delta\text{E}$  surface barrier and 5-mm E lithium drifted detectors. Deduced levels, L-transfers, J, and  $\pi$  from DWBA analysis.

 $^{34}\text{Cl}$  Levels

The experimental cross section is described as a sum over contributions from different angular momentum transfers, incorporating the spectroscopic factors  $A_L^2$ :  $(\text{d}\sigma/\text{d}\Omega)_\text{exp} = \alpha C \sum 1/(2J+1)[A_L^2(\text{d}\sigma/\text{d}\Omega)_\text{macro}^L + A_{L+2}^2(\text{d}\sigma/\text{d}\Omega)_\text{macro}^{L+2}]$ , where  $\alpha=1/3$  for  $(\text{d}, \alpha)$  reactions and  $\alpha=2J+1$  for  $(\alpha, \text{d})$  reactions on  $J_i=0$  nuclei.

| E(level) <sup>†</sup> | L <sup>†</sup> | Comments                                                                                                                                                                                                                                                                        |
|-----------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 152 10                | 2+4            | E(level): weighted average of 150 10 (1977Na10) and 180 40 (1982De29).<br>L: other: L=(2) from 1976De24.                                                                                                                                                                        |
| 463 10                | 0+2            | $A_{L=2}^2=0.71$ 16 for $3^+$ , $A_{L=4}^2<0.5$ for $3^+$ (1982De29).<br>E(level): weighted average of 460 10 (1977Na10) and 490 30 (1982De29).<br>L: other: L=2 from 1976De24.                                                                                                 |
| 1230 40               | 2              | $A_{L=0}^2=1.2$ 5 for $1^+$ , $A_{L=2}^2=0.3$ 3 (1982De29).<br>E(level): other: 1230 (1966Ki10).<br>$A_{L=2}^2=0.19$ 3 for $2^+$ (1982De29).                                                                                                                                    |
| 1882 10               |                | E(level): weighted average of 1880 10 (1977Na10) and 1920 50 (1966Ki10).                                                                                                                                                                                                        |
| 2177 7                | 2+4            | E(level): weighted average of 2180 10 (1977Na10) and 2175 7 (1982De29). Other: 2210 (1966Ki10).<br>$A_{L=2}^2=0.26$ 3 for $3^+$ , $A_{L=4}^2$ could not be determined (1982De29).                                                                                               |
| 2710 20               | 1+3            | E(level): weighted average of 2730 70 (1966Ki10) and 2708 20 (1982De29).<br>L: L=(3) from 1976De24.                                                                                                                                                                             |
| 3620 14               | 3+5            | $A_{L=1}^2=0.4$ 2 for $2^-$ , $A_{L=3}^2=0.24$ 15 for $2^-$ (1982De29).<br>E(level): weighted average of 3630 10 (1977Na10) and 3600 14 (1982De29).<br>L: others: L=5 from 1976De24 and 1977Na10.                                                                               |
| 4074 10               | 3+5            | $A_{L=3}^2=0.88$ 8 for $4^-$ , $A_{L=5}^2$ could not be determined (1982De29).<br>E(level): weighted average of 4080 10 (1977Na10) and 4056 18 (1982De29). Other: 4060 100 (1966Ki10).<br>L: others: L=(5) from 1976De24 and L=5 from 1977Na10.                                 |
| 4417 17               | 1,1+3          | $A_{L=3}^2=0.39$ 4 for $4^-$ , $A_{L=5}^2$ could not be determined (1982De29).                                                                                                                                                                                                  |
| 4678 10               | 1+3            | $A_{L=1}^2=2.6$ 5 for $0^-$ , $A_{L=1}^2=0.95$ 16 for $1^-$ , $A_{L=1}^2=0.4$ 3 for $2^-$ , $A_{L=3}^2<0.3$ (1982De29).<br>E(level): weighted average of 4680 10 (1977Na10) and 4650 40 (1982De29).                                                                             |
| 4788 10               | 1+3,3,4        | $A_{L=1}^2<0.8$ for $2^-$ , $A_{L=3}^2=0.6$ 3 for $2^-$ (1982De29).<br>E(level): weighted average of 4790 10 (1977Na10) and 4786 10 (1982De29). Other: 4790 (1966Ki10).<br>L: others: L=4 from 1977Na10, L=6 from 1976De24.                                                     |
| 4930 20               | 0+2,2          | $A_{L=3}^2=2.3$ 3 for $2^-$ , $A_{L=3}^2=1.48$ 15 for $3^-$ , $A_{L=4}^2=0.56$ 6 for $4^+$ (1982De29).<br>L: 2 is slightly preferred based on the DWBA fit (1982De29).<br>$A_{L=0}^2<2.14$ for $1^+$ , $A_{L=2}^2=1.4$ 10 for $1^+$ , $A_{L=2}^2=1.11$ 11 for $2^+$ (1982De29). |

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$^{32}\text{S}(\alpha, \text{d})$     **1982De29, 1977Na10, 1976De24 (continued)** $^{34}\text{Cl}$  Levels (continued)

| E(level) <sup>†</sup>                | L <sup>†</sup> | Comments                                                                                                                                                                                                            |
|--------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5279 10                              | 6              | E(level): weighted average of 5280 10 (1977Na10) and 5277 10 (1982De29). Others: 5230 (1966Ki10), 5283 (1976De24).<br>L: 6 from 1976De24, 1977Na10, and 1982De29.<br>$A^2_{L=6}=1.0$ for 7 <sup>+</sup> (1982De29). |
| 5.85×10 <sup>3</sup> <sup>‡</sup> 15 |                |                                                                                                                                                                                                                     |
| 6300 <sup>‡</sup>                    |                |                                                                                                                                                                                                                     |
| 6540 <sup>‡</sup>                    |                |                                                                                                                                                                                                                     |
| 6950 <sup>‡</sup>                    |                |                                                                                                                                                                                                                     |
| 7350 20                              | 3,6            | E(level): other: 7370 (1966Ki10).<br>$A^2_{L=3}=1.47$ 13 for 3 <sup>-</sup> , $A^2_{L=6}=0.60$ 5 for 6 <sup>+</sup> , $A^2_{L=6}=0.34$ 3 for 7 <sup>+</sup> (1982De29).                                             |
| 7840 30                              | 3,5,6          | $A^2_{L=3}=1.08$ 11 for 3 <sup>-</sup> , $A^2_{L=5}=0.65$ 6 for 5 <sup>-</sup> , $A^2_{L=6}=0.41$ 4 for 6 <sup>+</sup> , $A^2_{L=6}=0.23$ 3 for 7 <sup>+</sup> (1982De29).                                          |
| 8390 20                              | 3,5,6          | $A^2_{L=3}=1.86$ 17 for 3 <sup>-</sup> , $A^2_{L=5}=1.05$ 9 for 5 <sup>-</sup> , $A^2_{L=6}=0.64$ 6 for 6 <sup>+</sup> , $A^2_{L=6}=0.36$ 4 for 7 <sup>+</sup> (1982De29).                                          |

<sup>†</sup> From 1982De29, unless otherwise noted.<sup>‡</sup> From 1966Ki10.